



Figure 3: OpenXenManager



Figure 4: Citrix XenCenter

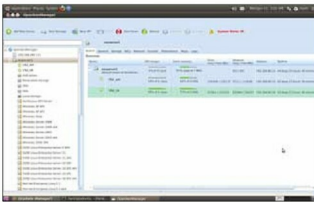


Figure 5: XenServer added

high end to the low end, but for our purposes, AMD Phenom II X6 with six-core will suit us the best. Another factor that impacts successful virtualisation is RAM—you must populate all the available DIMMs with ample RAM on your motherboard, 8 GB RAM would be good and 16 GB great. Don't forget to add storage—1 TB HDD will be sufficient for all your guest virtual machines and you also require a laptop or desktop to install OpenXenManager (for Linux) or Citrix XenCenter (for PC) to manage your virtual home lab.

Hypervisor installation

XenServer requires virtualisation extensions (VT-x or AMD-V) to run HVM (Hardware Virtual Machine) mode VMs (e.g., any Windows VM), and without them you can only run PV (ParaVirtualization) mode VMs (e.g., any guest OS modified to run on a hypervisor). First download Citrix XenServer from www.citrix.com/xenserver, and burn it on a CD to directly install on your hardware. Use the CD as the first boot device. The installation of XenServer is straightforward and resembles



Figure 6: New storage repositories



Figure 7: A new VM

the text-based installation of Linux.

During installation, provide one static IP to XenServer for management and also remember the root user password provided, as it is required to connect with the management console. When the installation finishes, the machine will reboot and welcome you with the Citrix XenServer splash screen. Citrix XenServer 6.0 hypervisor provides console-based access for further configuring the hypervisor using OpenXenManager.

OpenXenManager installation

Here, I'm not using Citrix XenCenter for reasons related to FOSS freedom. OpenXenManager is an open source clone of Citrix XenCenter, earlier called open XenCenter. It is based on Python and requires `python-gtk-vnc` (`sudo apt-get install python-gtk-vnc`) libraries to be installed first, as it communicates with XenServer and the VMs inside it using VNC. I'm using Ubuntu 10.04 to install OpenXenManager. The installation of OpenXenManager is very simple. Go to <http://sourceforge.net/projects/openxenmanager/files/> and download the rev 48 tarball, untar the downloaded file to your preferred path for software installs and open the terminal. Then go to the untar location, type `python window.py` and OpenXenManager will appear to serve you. Click on *add new server* icon and a dialogue box will appear asking for the IP and root user password for the connection provided during hypervisor installation. After successful connection, XenServer will appear.